

Introduction to survival analysis

Kaplan-Meier analysis, risk modeling

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Learning objectives and expected competencies

This course aims to provide students with both a rigorous and practical understanding of survival models and their statistical implementation. By the end of the module, students should be able to analyze, interpret, and compare different models depending on the characteristics of the available data.

1. Understanding the context and motivations

Classical regression or classification methods are not suitable for survival data because they assume that the response variable is always fully observed. However, in survival settings, some observations are censored: the exact time of occurrence of the event is unknown; we only know that it exceeds a certain duration. Excluding such individuals would result in the loss of partial but valuable information and would introduce bias into the analysis. Moreover, the event risk evolves over time and cannot be treated as a simple fixed variable. Survival analysis therefore aims to model the probability that an individual “survives” beyond a given time point while accounting for both observed events and censored observations, which the Kaplan–Meier estimator and the Cox model allow us to handle rigorously.

- Identify situations where a **survival analysis** approach is appropriate: time until occurrence of an event (death, relapse, failure, defect, etc.), often in the presence of **censored data**.
- Explain why a classical regression or classification approach is not appropriate for this type of data (unobserved event times, information asymmetry, time dependence).
- Understand the fundamental concepts: survival function $S(t)$, hazard function $h(t)$, and the relationship between them.

2. Mastering descriptive methods: Kaplan–Meier

- Be able, from a dataset, to:
 1. estimate the empirical survival curve using the Kaplan–Meier estimator;
 2. plot the curve $\hat{S}_{KM}(t)$ together with its confidence intervals (Greenwood);
 3. interpret graphically the differences between groups.
- Apply and interpret the **log-rank test** to compare two (or more) survival functions, explicitly stating the underlying assumptions (non-informative censoring, independence of subjects, proportional hazards).

3. Modeling with covariates: Cox model

- Build a **proportional hazards Cox model** using explanatory covariates.
- Understand that the model is a *hazard-based model*, whose main objective is to describe the influence of covariates on the instantaneous risk rather than to predict the exact event time T .

- Know how to interpret the estimated coefficients:

$$\hat{\beta}_j > 0 \Rightarrow X_j \text{ increases the risk,} \quad \hat{\beta}_j < 0 \Rightarrow X_j \text{ has a protective effect.}$$

- Construct and interpret **individual risk scores** $\eta_i = \mathbb{X}_i^\top \hat{\beta}$, then perform **risk stratification** (low-, medium-, or high-risk groups).
- Relate this stratification to hypothetical therapeutic decisions or differentiated follow-up strategies.
- Evaluate the quality of the model using:
 - **discrimination** metrics (C-index, AUC(t));
 - **calibration** metrics (Brier score, calibration plot).

4. Critical reading and comparison of approaches

Aspect	Kaplan–Meier	Cox model
Model type	Nonparametric (no assumption on the form of $\mathcal{S}(t)$)	Semiparametric: separates $h_0(t)$ and $\exp(\mathbb{X}^\top \beta)$
Use of covariates	No explanatory covariates	Incorporates explanatory covariates
Main objective	Describe the empirical survival of a group	Quantify the effect of covariates on instantaneous risk
Interpretation	Graphical interpretation (comparison between groups)	Interpretation via hazard ratios $\exp(\beta_j)$
Main advantage	Simple, robust, intuitive	Flexible, allows stratification and relative prediction
Main limitations	Does not allow adjustment for multiple variables	Proportional hazards assumption, more abstract interpretation

Table 1: Comparison between Kaplan–Meier and the Cox model.

5. Expected competencies

At the end of the course, students should be able to:

- identify the model appropriate to the data context (KM vs Cox);
- implement the methods in `Python` or `R` (with packages `lifelines`, `sksurv`, `survival`);
- correctly interpret model outputs and justify the decisions made;
- clearly present and justify a complete survival analysis: estimation, testing, interpretation, validation.

The ultimate objective is not only to apply a model, but to understand the *statistical reasoning* that connects the observed data, the estimated survival function, and risk-based decision-making.

Chapter 1

Fundamental notions: random variable, trajectory, and distribution

This chapter introduces the fundamental probability concepts required to formalize survival models. Before addressing risk modeling, it is important to understand what a random variable is, what an observation (or trajectory) represents, and how the probability distribution describes the structure of randomness. These concepts will serve as the foundation for the entire statistical framework presented in the following chapters.

1.1 Random variable: from randomness to a measurable quantity

Definition 1.1.1 (Random variable). A *random variable* is a numerical quantity whose value depends on randomness. In other words, it is a mapping that associates each possible outcome of a random experiment with a real number.

Formally, we write:

$$\begin{aligned} T : \Omega &\longrightarrow \mathbb{R} \\ \omega &\longmapsto T(\omega) \end{aligned}$$

where:

- Ω denotes the set of possible outcomes of the experiment (called the *sample space*);
- $\omega \in \Omega$ is a particular outcome (for example, a specific individual or an experimental scenario);
- $T(\omega)$ is the numerical value observed for this outcome (for example, the time until failure for that individual).

Thus, T transforms the random outcome ω into a measurable and interpretable real value.

The symbol Ω represents the “world of possible outcomes”: all situations that could occur. The random variable T is a *function of randomness*: before selecting an outcome ω , its value is unknown. Once the experiment has been carried out (a trajectory observed), $T(\omega)$ becomes a concrete value, namely the one observed in the data. In survival analysis, T denotes the random time until the event of interest (death, relapse, failure, etc.), and each subject in the cohort corresponds to one outcome ω_i .

Example. Suppose we observe the time until failure of a device. The random variable T associates

with each device the duration (in days, months, etc.) until failure occurs. Before performing the experiment, the value of T is unknown, but it may take certain possible values according to a probabilistic mechanism. The key idea is that T is not a fixed value, but a variable whose outcome is uncertain. It models randomness numerically: duration, height, score, number of trials, etc.

After introducing the concept of a random variable, we now examine what a *concrete observation* of this variable represents, namely a trajectory.

1.2 Trajectory or observation: the concrete realization of randomness

Definition 1.2.1 (Trajectory or observation). A *trajectory* (or *realization*) of a random variable corresponds to the value actually observed for this variable for a particular outcome of randomness.

Formally, if $T : \Omega \rightarrow \mathbb{R}$ is a random variable, then for a given outcome $\omega \in \Omega$, the quantity

$$T(\omega)$$

denotes the *trajectory* (or the *realized value*) of T associated with this outcome ω .

In other words, $T(\omega)$ is the concrete value taken by the random variable T when the experiment produces the outcome ω .

Each outcome ω in the sample space Ω corresponds to a particular individual, device, or experimental scenario. Thus, when observing a sample of n subjects, we actually observe n trajectories:

$$T(\omega_1), T(\omega_2), \dots, T(\omega_n),$$

which are often denoted $T_1, T_2, \dots, T_n$.

For example, if T represents the time until device failure, an observation $T_i = 7.3$ means that for the i -th device (corresponding to outcome ω_i), failure occurred after 7.3 days. Each observed subject therefore provides a particular trajectory of the global random phenomenon modeled by T .

The random variable represents the *abstract model of randomness*, whereas a trajectory corresponds to a *concrete data point*. In other words: T is a theoretical concept, whereas T_i is an observation in a dataset. In statistics, we observe several realizations $T_1, T_2, \dots, T_n$ of the same random variable T in order to estimate its characteristics (mean, median, distribution, etc.).

We now understand the distinction between a random variable (theoretical concept) and its observation (empirical value). It remains to understand how the *probability distribution* mathematically describes how these values are distributed.

1.3 Probability distribution: describing the behavior of randomness

Definition 1.3.1 (Probability distribution of a real-valued random variable). The *distribution* of a random variable describes how its possible values are distributed over the real numbers, together with their associated probabilities.

For a continuous random variable T , this distribution is described by:

- a **cumulative distribution function** $F(t) = \mathbb{P}(T \leq t)$, which gives the probability that T takes a value less than or equal to t ;
- a **survival function** $S(t) = 1 - F(t) = \mathbb{P}(T > t)$, useful in survival analysis;
- a **probability density function** $f(t)$ such that $F'(t) = f(t)$ for almost every t .

The distribution of a random variable is to randomness what a *map of a territory* is to geography: it indicates which values are more or less likely. In survival analysis, the distribution of T describes how possible lifetimes are distributed within the population: some failures occur very early, others very late, and this distribution is characterized by $\mathcal{S}$, f , or h .

The standard distributions that we will study (exponential, Weibull, log-normal, Cox, etc.) specify particular functional forms for $\mathcal{S}$ and h .

1.4 Maximum likelihood principle

1.4.1 Motivation and intuition

One of the foundations of statistical inference consists of *estimating unknown parameters* from an observed sample. The idea behind **maximum likelihood estimation** (denoted *MLE*) is conceptually very simple: we seek the parameter value that makes the observed data the most “likely”.

In other words, among all possible models, we choose the one for which the actually observed data would have had the greatest chance of occurring. This idea can be interpreted as a form of *empirical coherence principle*: the model should best explain what has been observed.

Example 1.4.1 (Intuition). Suppose we believe that a variable Y follows a normal distribution $\mathcal{N}(\mu, \sigma^2)$. The parameters μ and σ^2 are unknown. The likelihood principle consists of finding the values of μ and σ that make the observed sample $(y_1, \dots, y_n)$ most probable under this normality assumption. Maximum likelihood estimation therefore selects the parameters $(\hat{\mu}, \hat{\sigma})$ that maximize the joint probability of the observed data.

The maximum likelihood principle relies on a form of “inverse reasoning”: we start from the observations and search for the most plausible model that could have generated them. This approach is universal: whether dealing with lifetimes, counts, survival probabilities, or risk scores, the idea is always the same — finding the parameters that make the data most likely.

1.4.2 Mathematical formulation

Let an independent and identically distributed (i.i.d.) random sample be given by

$$Y_1, Y_2, \dots, Y_n \sim f(y; \theta),$$

where $f(y; \theta)$ denotes the density (or probability) function parameterized by a vector $\theta \in \Theta \subset \mathbb{R}^p$.

Definition 1.4.1 (Likelihood function). The *likelihood function* is defined as

$$L(\theta) = \prod_{i=1}^n f(y_i; \theta),$$

and represents the probability (or density) of observing the data $(y_1, \dots, y_n)$ under the model parameterized by θ .

Definition 1.4.2 (Maximum likelihood estimator). The maximum likelihood estimator (MLE) is defined as

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} L(\theta) = \arg \max_{\theta \in \Theta} \log L(\theta).$$

Remark 1 (Log-transformation). In practice, we almost always work with the *log-likelihood*:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(y_i; \theta),$$

because the logarithm transforms products into sums (which are easier to manipulate), without changing the location of the maximum.

Propriété 1.4.1 (Maximum condition). *Under regularity conditions, the estimator $\hat{\theta}_{\text{MLE}}$ satisfies*

$$\left. \frac{\partial \ell(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}} = 0, \quad \text{and} \quad \left. \frac{\partial^2 \ell(\theta)}{\partial \theta^2} \right|_{\theta=\hat{\theta}} < 0,$$

that is, the gradient of the log-likelihood vanishes at the maximum point, and the Hessian matrix there is negative definite.

1.4.3 Statistical properties and relevance

Propriété 1.4.2 (Asymptotic properties). *Under general assumptions, the maximum likelihood estimator has remarkable properties:*

- **Consistency:** $\hat{\theta}_{\text{MLE}} \rightarrow \theta^*$ as $n \rightarrow \infty$;
- **Asymptotic normality:**

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta^*) \xrightarrow{\mathcal{L}} \mathcal{N}(0, \mathcal{I}(\theta^*)^{-1}),$$

where $\mathcal{I}(\theta)$ denotes the Fisher information matrix, which will be defined later in equation (8.2);

- **Asymptotic efficiency:** *among regular estimators, the MLE achieves the minimum possible variance given by the Cramér–Rao lower bound.*

Remark 2 (Geometric interpretation). Maximum likelihood estimation can be interpreted as searching for the parameter value θ that best matches the empirical shape of the data distribution. In other words, we adjust the parametric curve $f(y; \theta)$ so that it lies as close as possible to the observed empirical distribution. This geometric interpretation is very useful for understanding why the MLE converges toward the true parameter as n increases.

1.4.4 Connection with survival analysis

The likelihood principle plays a central role in survival analysis. In this setting, the likelihood function must account for both:

- complete observations (subjects who experienced the event, contributing density terms $f(t_i)$);
- censored observations (subjects who left the study before the event, contributing survival terms $S(t_i)$).

Thus, the total likelihood often takes the form

$$L(\theta) = \prod_{i=1}^n [f(t_i; \theta)]^{\delta_i} [S(t_i; \theta)]^{1-\delta_i},$$

where $\delta_i = \mathbb{1}_{\{T_i \text{ observed}\}}$ indicates whether the event was observed. This principle will be revisited and adapted in the Cox model, where only the *partial likelihood* (without the baseline hazard) is used to estimate the effects of covariates.

1.5 Ridge regression: formulation, solution, and interpretation

1.5.1 Design matrix and reminder of the linear model

We consider a standard linear model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where:

- $\mathbf{y} \in \mathbb{R}^n$ is the vector of observed responses;
- $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the **design matrix** (or **covariate matrix**), where each row corresponds to one observation and each column to one explanatory variable;
- $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of unknown coefficients to be estimated;
- $\boldsymbol{\varepsilon}$ is the error vector (assumed to be centered and to have variance σ^2).

1.5.2 Ordinary least squares (OLS) estimator

The ordinary least squares estimator is obtained by minimizing the sum of squared residuals:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

Under the assumption that $\mathbf{X}^\top \mathbf{X}$ is invertible, we obtain the closed-form solution:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}, \quad \text{and} \quad \hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}}_{\text{OLS}} = \underbrace{\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top}_{\mathbf{H}} \mathbf{y},$$

where $\mathbf{H}$ is called the **hat matrix**.

1.5.3 Ridge regression: formulation of the problem

The Ridge model introduces a quadratic penalty on the norm of the coefficients:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \arg \min_{\boldsymbol{\beta}} (\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2), \quad \lambda \geq 0.$$

The parameter λ controls the strength of the regularization:

- if $\lambda = 0$, we recover the standard OLS solution;
- if $\lambda \rightarrow +\infty$, the coefficients are heavily penalized and tend toward 0.

1.5.4 Derivation of the closed-form solution

We seek to set the gradient of the objective function equal to zero:

$$\mathcal{L}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2.$$

Step 1. Compute the gradient.

$$\nabla_{\boldsymbol{\beta}} \mathcal{L}(\boldsymbol{\beta}) = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\lambda \boldsymbol{\beta}.$$

Step 2. First-order condition.

$$-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\lambda \boldsymbol{\beta} = \mathbf{0}.$$

Step 3. Simplification.

$$(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})\boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}.$$

Step 4. Explicit solution.

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}.$$

It follows that:

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}}_{\text{Ridge}} = \underbrace{\mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top}_{\mathbf{S}_\lambda} \mathbf{y},$$

where $\mathbf{S}_\lambda$ is called the **smoother matrix** or the **Ridge hat matrix**.

1.5.5 Spectral interpretation and numerical stabilization

Using the singular value decomposition (SVD)

$$\mathbf{X} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top,$$

with $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_p)$, we obtain:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \mathbf{V} \text{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda}\right) \mathbf{U}^\top \mathbf{y}.$$

Each principal direction is *shrunk* by the factor

$$\frac{\sigma_i}{\sigma_i^2 + \lambda} \in (0, 1].$$

Thus:

- well-informed directions (large σ_i) are only slightly affected;
- unstable directions (small σ_i) are strongly damped, which pulls the corresponding coefficients toward 0.

Practical consequences:

- the matrix $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ is always invertible (better conditioning);
- the variance of the estimators is reduced;
- the analysis is stabilized, especially in the presence of multicollinearity or noisy data.

Remark. Ridge regression introduces a small bias, but it often improves the model's *predictive accuracy* out of sample through a better bias–variance trade-off.

Chapter 2

Motivations and formal introduction

This chapter sets out the conceptual framework and motivations of survival analysis. We begin with concrete examples (*device failure*, *cancer relapse*, *death within a cohort*) in order to introduce the central mathematical objects: the event time T , the censoring time C , the observed time $Y = \min(T, C)$, the event indicator $D = \mathbb{1}_{\{T \leq C\}}$, the survival function $\mathcal{S}$, the instantaneous hazard (or *hazard rate*) h , and the cumulative hazard H . To give a few concrete examples motivating such modeling and its use in everyday applied settings:

- Failure of an electronic device: We follow a batch of n hard drives from the time they enter service. The event of interest is the first irreversible failure. Some drives are still operating at the end of the study, so their failure times are *right-censored*. Typical objectives are to estimate reliability beyond a given horizon, compare brands or usage conditions, and predict short-term failure risk for maintenance planning.
- Risk of cancer relapse: For patients treated for cancer, the event of interest is the first relapse. Subjects who do not relapse before the end of follow-up (or who are lost to follow-up) are censored. The aim is to estimate *relapse-free survival*, compare therapeutic strategies, and quantify the effect of covariates (age, stage, biomarkers).
- Death within a cohort: In a population-based cohort, the event is death (all-cause or cause-specific). We study overall survival, the influence of risk factors, and report interpretable quantities (median survival, instantaneous mortality rate).

What we call a **cohort** is the set of subjects (individuals, devices, biological units, etc.) followed over time according to explicit inclusion and follow-up rules, in order to observe the occurrence of an event of interest. Each subject is described by baseline covariates (e.g., age, sex, technical characteristics) and possibly by covariates that evolve over time.

2.1 Survival study: operational definitions

Definition 2.1.1 (Survival study). A *survival study* specifies the following elements:

- (i) a **source population**;
- (ii) **inclusion and exclusion criteria**;
- (iii) a **time origin** (e.g., entry into service, diagnosis, randomization);
- (iv) an **event of interest** (e.g., failure, relapse, death);
- (v) an **observation period** $[t_0, t_1]$;

(vi) **censoring** rules (loss to follow-up, end of follow-up).

The random *event time* is denoted by T .

Remark 3 (Time origin and time scales). The choice of the time origin (age, time since inclusion, calendar time) is crucial: it affects the interpretation of the parameters and the validity of comparisons. We will discuss common time scales and delayed entry later on.

2.2 Observed variables and notation

Definition 2.2.1 (Elementary data with censoring). Let $T \in [0, \infty]$ be the (continuous) time to the event, and $C \in [0, \infty]$ the censoring time. We *observe*

$$Y = \min(T, C), \quad D = \mathbb{1}_{\{T \leq C\}},$$

together with covariates $X \in \mathcal{X}$ (fixed or time-dependent). For $i = 1, \dots, n$, we denote by (X_i, Y_i, D_i) the individual observations, assumed to be independent or conditionally independent.

The key idea is that, in a survival study, we do not always know T . If the event has not yet occurred by the end of follow-up (or if the subject is lost to follow-up), we only know that $T > C$. This is called *right-censoring*. The variable Y therefore corresponds to the *observed* duration, and D indicates whether the event was actually observed ($D = 1$) or censored ($D = 0$).

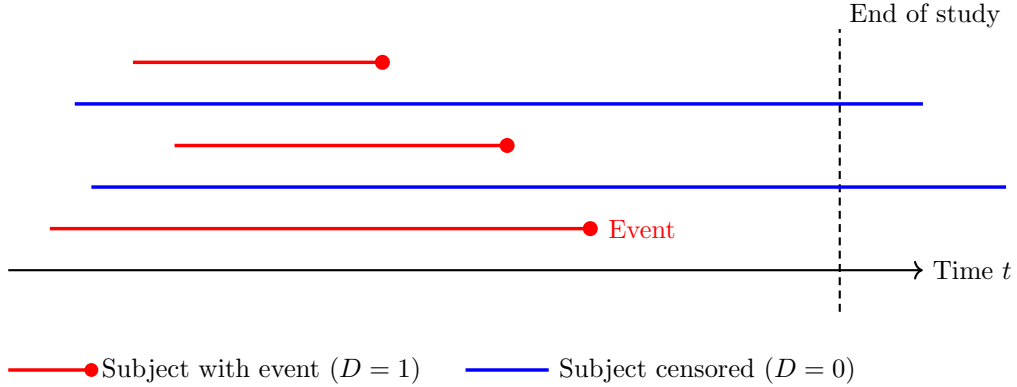


Figure 2.1: Schematic representation of follow-up times in a survival study. The red segments indicate subjects who experienced the event ($D = 1$), whereas the blue segments represent censored subjects ($D = 0$). The dotted vertical line marks the end of the study.

The example above illustrates an essential feature of survival data: not all subjects necessarily experience the event of interest before the end of the study. Some have their follow-up interrupted prematurely (change of hospital during medical follow-up, end of observation, treatment discontinuation, etc.); this is precisely what *censoring* means.

This makes standard “regression-type” modeling inappropriate, since the complete target variable T_i is not always available for every individual. Nevertheless, censored subjects contain valuable information: the mere fact that they survived up to a certain time contributes to our understanding of the phenomenon under study. It would therefore be unfortunate, and even statistically biased, to exclude them.

This is precisely the strength of survival analysis: it makes it possible to jointly exploit complete observations (events) and incomplete observations (censored cases), by incorporating their respective contributions into an appropriate likelihood function. In other words, even an individual who has not yet experienced the event still contributes to statistical inference, because we know that they have “survived” at least until their censoring time.

The modeling of *survival* thus relies on a simple but powerful idea: learning how to describe, estimate, and predict the behavior of risk over time, while respecting the partially observed nature of the data. This formalism will form the basis of the models studied in the following chapters—whether parametric, semiparametric (such as the Cox model), or modern (neural survival models).

Definition 2.2.2 (Survival function and associated functions). The *survival function* of T is

$$\mathcal{S}(t) = \mathbb{P}(T > t), \quad t \geq 0.$$

The cumulative distribution function is $F(t) = \mathbb{P}(T \leq t) = 1 - \mathcal{S}(t)$. If T admits a density f , then $f(t) = F'(t)$ *almost everywhere*.

The function $\mathcal{S}(t)$ gives the probability of *surviving beyond time t* . It decreases from 1 (at time 0) toward 0 (in the long run). Graphically, it represents the proportion of subjects still “alive” over time. By contrast, the derivative $f(t)$ represents the *rate at which events occur* around time t . These two notions, $\mathcal{S}$ and f , are fundamental and will underlie all later modeling.

Definition 2.2.3 (Instantaneous hazard and cumulative hazard). The *instantaneous hazard* (or *hazard rate*) is defined by

$$h(t) = \lim_{dt \rightarrow 0} \frac{\mathbb{P}(t \leq T < t + dt \mid T \geq t)}{dt}, \quad (t > 0).$$

The associated *cumulative hazard* is

$$H(t) = \int_0^t h(u) du.$$

The hazard rate $h(t)$ measures the instantaneous probability that an event occurs at time t , *given* that the subject has survived up to that time. This infinitesimal approach is in fact very intuitive: we examine the risk that the event occurs in a time interval dt immediately after t , for a subject still under observation at that moment.

It is therefore an *instantaneous conditional risk*, and not a direct probability observed over a finite duration. By integrating this instantaneous risk over time, we obtain the quantity $H(t)$, which accumulates exposure to danger from the beginning of follow-up up to time t .

Since $h(t)$ describes risk over an infinitesimal interval, the integral $\int_0^t h(u) du$ can be interpreted as the *continuous sum* of these micro-risks over time. Thus, $H(t)$ has a concrete interpretation: it represents the *total accumulated risk* to which an individual has been exposed between the beginning of the study and time t .

Théorème 2.2.1 (Fundamental relationship between $\mathcal{S}$, h , and H). *If h is locally integrable, then for every $t \geq 0$:*

$$\mathcal{S}(t) = \exp(-H(t)) = \exp\left(-\int_0^t h(u) du\right), \quad f(t) = h(t) \mathcal{S}(t).$$

Proof. By definition of the instantaneous hazard:

$$\begin{aligned}
h(t) &\underset{dt \rightarrow 0}{=} \frac{\mathbb{P}(t \leq T < t + dt \mid T > t)}{dt} + o(1) && \text{(definition of } h(t)) \\
&\underset{dt \rightarrow 0}{=} \frac{1}{\mathbb{P}(T \geq t)} \frac{\mathbb{P}(\{t \leq T < t + dt\} \cap \{T \geq t\})}{dt} + o(1) && \text{(Bayes' formula)} \\
&\underset{dt \rightarrow 0}{=} \frac{1}{\mathcal{S}(t)} \frac{\mathbb{P}(t \leq T < t + dt)}{dt} + o(1) && \text{(since } \mathcal{S}(t) = \mathbb{P}(T \geq t)) \\
&\underset{dt \rightarrow 0}{=} \frac{1}{\mathcal{S}(t)} \frac{\mathbb{P}(T \leq t + dt) - \mathbb{P}(T \leq t)}{dt} + o(1) && \text{(property of } F(t)) \\
&\underset{dt \rightarrow 0}{=} -\frac{1}{\mathcal{S}(t)} \frac{\mathcal{S}(t + dt) - \mathcal{S}(t)}{dt} + o(1) && \text{(since } F = 1 - \mathcal{S}).
\end{aligned}$$

Taking the limit as $dt \rightarrow 0$, we obtain:

$$h(t) = -\frac{\mathcal{S}'(t)}{\mathcal{S}(t)} \iff \mathcal{S}'(t) + h(t)\mathcal{S}(t) = 0.$$

This differential equation is solved by integration:

$$\mathcal{S}(t) = \exp\left(-\int_0^t h(s) ds\right).$$

We then define the *cumulative hazard* $H(t) = \int_0^t h(s) ds$, which yields the relation:

$$\mathcal{S}(t) = e^{-H(t)}.$$

Finally, since $f(t) = F'(t) = -\mathcal{S}'(t)$, we indeed have:

$$f(t) = h(t)\mathcal{S}(t),$$

which concludes the proof. □

2.3 Types of censoring and truncation

Definition 2.3.1. We define the notions of right-, left-, and interval-censoring:

- **Right-censoring:** we know that $T > Y$ when $D = 0$ (end of study, loss to follow-up).
- **Left-censoring:** we only know that $T \leq Y$ (e.g., seroconversion before the first positive test).
- **Interval-censoring:** we know that $T \in (L, R]$ (e.g., periodic check-ups).

Definition 2.3.2. We also define the notion of truncation:

- **Left truncation / delayed entry:** a subject enters the cohort only if $T > U$ (inclusion after the time origin), which conditions the observable risk.
- **Right truncation:** only subjects with $T \leq V$ are observable (rarer in practice in classical survival analysis).

Remark 4 (Non-informative censoring). Many methods assume *independent censoring* (*non-informative censoring*), typically $T \perp\!\!\!\perp C$ (or conditional independence given X). This assumption guarantees the consistency of estimators such as Kaplan–Meier. Its violation requires dedicated approaches (IPCW, joint models, etc.).

2.4 Dynamic objects: risk set and counting process

Definition 2.4.1 (Risk set). At time t , the *risk set* is

$$\mathcal{R}(t) = \{i \in \{1, \dots, n\}, Y_i \leq t\}.$$

It is therefore the set of subjects still in the study at time t and represents the population effectively *at risk* at time t .

Definition 2.4.2 (Counting process N and exposure process Y). The data can be represented by a counting process

$$N(t) = \sum_{i=1}^n \mathbb{1}_{\{T_i \leq t, D_i=1\}}$$

and an exposure process

$$Y(t) = \sum_{i=1}^n \mathbb{1}_{\{i \in \mathcal{R}(t)\}}.$$

These objects will be useful for nonparametric estimators and semiparametric models.

2.5 Quantities of interest and interpretations

Definition 2.5.1. Here is a non-exhaustive list of important quantities in survival analysis:

- **Median survival:** $\text{Med}(T) = \inf\{t : S(t) \leq 1/2\}$.
- **Survival at a horizon t^* :** $S(t^*)$.
- **Conditional hazard function:** $t \mapsto h(t \mid \mathbb{X})$ (when covariates are considered).
- **Metrics in survival analysis:** e.g., *C-index*, *td-AUC*, *Brier Score*, ...

These different quantities make it possible to approach survival from several complementary angles. The *median survival* provides a synthetic and robust measure: it indicates the time beyond which 50% of subjects are still alive.

When covariates $\mathbb{X}$ are introduced, the function $h(t \mid \mathbb{X})$ describes how individual characteristics modify the risk over time. Finally, model quality measures (such as the C-index or the Brier Score) make it possible to assess the *predictive quality* of a survival model: they measure its ability to rank subjects correctly according to their risk and to estimate their survival probabilities accurately.

Thus, survival analysis is not limited to estimating a single curve: it aims at a refined understanding of risk over time, its determinants, and the performance of the models used to describe it.

2.6 Basic statistical assumptions

In a properly designed cohort, one generally assumes:

- (i) independent subjects (or conditionally independent subjects);
- (ii) **non-informative** censoring, that is,

$$T \perp\!\!\!\perp C \mid \mathbb{X},$$

where $\mathbb{X}$ denotes the set of observed covariates;

- (iii) **error-free** recording of key dates (event, censoring, inclusion, etc.).

These assumptions form the basis of the classical framework of survival analysis, but they may be discussed and relaxed when the empirical context requires it.

Remark 5 (Internal/external validity and bias). Results depend on the study design (selection, loss to follow-up, inclusion delays). One must be careful about calendar-time biases, *immortal time bias*, and unmeasured confounding. External validity requires that the source population and follow-up conditions be comparable to the application context.

Chapter 3

Some standard distributions in survival analysis

The objective of this chapter is to present the main probability distributions used to model survival times. These distributions are fundamental building blocks: they make it possible to express explicitly the survival function $\mathcal{S}(t)$, the instantaneous hazard $h(t)$, and the cumulative hazard $H(t)$. They serve as the basis for parametric models and, more generally, for understanding the behavior of risk over time.

3.1 The exponential distribution

Definition 3.1.1 (Exponential distribution). A random variable T follows an *exponential distribution* with parameter $h > 0$, denoted $T \sim \text{Exp}(h)$, if its density is

$$f(t) = he^{-ht}, \quad t \geq 0.$$

Its survival function and instantaneous hazard are:

$$\mathcal{S}(t) = e^{-ht}, \quad h(t) = h.$$

The exponential model is the simplest possible model in survival analysis: the hazard rate $h(t)$ is constant. In other words, the instantaneous probability of an event does not depend on the elapsed time. This amounts to saying that the system does not “age”: the risk remains the same at every point in follow-up.

Propriété 3.1.1 (Memoryless property). *The exponential distribution (and its discrete analogue, the geometric distribution) satisfies the memoryless property:*

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t), \quad s, t \geq 0, \quad T \sim \text{Exp}(h).$$

The *memoryless* character means that the probability of surviving for an additional t units of time does not depend on the time already elapsed. This property is intuitive in some contexts:

- **Electronic devices:** if the failure of a component depends only on a random shock (and not on wear), the fact that it has already operated for a long time does not influence its immediate probability of failure.
- **Chronic diseases:** in some stable pathologies, the risk of relapse may be considered constant over time, as long as no major biological state change occurs.

However, this assumption is often too simplistic: most real phenomena exhibit a risk that *varies* over time; hence the interest of more flexible distributions such as Weibull or log-normal.

3.2 The Weibull distribution

Definition 3.2.1 (Weibull distribution). A random variable T follows a *Weibull distribution* with parameters (h, k) , where $h > 0$ and $k > 0$, denoted $T \sim \text{Weibull}(k, h)$, if its density is:

$$f(t) = \frac{k}{h} \left(\frac{t}{h} \right)^{k-1} e^{-(t/h)^k}, \quad t \geq 0.$$

The survival function and instantaneous hazard are:

$$\mathcal{S}(t) = e^{-(t/h)^k}, \quad h(t) = \frac{k}{h} \left(\frac{t}{h} \right)^{k-1}.$$

Remark 6. The Weibull distribution is a *natural generalization* of the exponential distribution:

- If $k = 1$, we recover the exponential distribution with parameter h^{-1} .
- If $k > 1$, the hazard $h(t)$ *increases over time*: this models an aging or wear-out phenomenon.
- If $k < 1$, the hazard *decreases over time*: the danger is higher at the beginning, then declines.

This is why this distribution is ubiquitous in industrial applications (material reliability, component lifetime) and medical applications (evolution of risk over time since treatment or surgery). It offers an ideal compromise between analytical simplicity and flexibility of the hazard profile.

3.3 Other useful parametric distributions

Definition 3.3.1 (Log-normal distribution). A random variable T follows a *log-normal* distribution if $\log(T)$ follows a normal distribution with mean μ and standard deviation $\sigma > 0$. We then write $T \sim \text{LogNormal}(\mu, \sigma)$, with density:

$$f(t) = \frac{1}{t\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log t - \mu)^2}{2\sigma^2}\right), \quad t > 0.$$

The log-normal distribution models well phenomena whose durations result from the *multiplication* of several independent factors (as opposed to addition for the normal distribution). Its hazard $h(t)$ is often bell-shaped: low at the beginning, maximal at some point in time, then decreasing. It is a good choice for diseases whose relapse risk is highest during a critical period after treatment, then declines.

Definition 3.3.2 (Log-logistic distribution). $T \sim \text{LogLogistic}(\alpha, \beta)$ if

$$\mathcal{S}(t) = \frac{1}{1 + (t/\alpha)^\beta}, \quad t > 0.$$

The instantaneous hazard is:

$$h(t) = \frac{\frac{\beta}{\alpha}(t/\alpha)^{\beta-1}}{1 + (t/\alpha)^\beta}.$$

This distribution is widely used for its analytical simplicity and its ability to generate non-monotonic hazard profiles (increasing then decreasing). It also allows closed-form expressions for the median and the survival function, which facilitates interpretation and estimation.

Chapter 4

Modeling the survival function: the Kaplan–Meier approach

The survival function $\mathcal{S}(t) = \mathbb{P}(T > t)$ describes the probability that an individual or a system has not yet experienced the event of interest at time t . In the previous chapters, we saw how $\mathcal{S}$ can be interpreted and how some parametric distributions (exponential, Weibull, etc.) provide analytical forms for it.

However, in many practical situations, one does not wish to impose a specific form on the distribution of T . One then seeks to *estimate empirically* the survival function from the observed data (Y_i, D_i) , while taking censoring into account. The Kaplan–Meier estimator [KM58], also called the *product-limit estimator*, is the standard method for this task.

4.1 Principle and intuition

The idea of Kaplan–Meier (1958) is based on a simple and intuitive approach: we estimate the probability of surviving up to time t as the *product* of the probabilities of surviving at each time at which an event was observed.

In other words, instead of modeling a density $f(t)$ or a hazard $h(t)$, we focus directly on the empirical survival probability:

$$\mathcal{S}(t) = \mathbb{P}(T > t).$$

Each time an event (death, failure, relapse, etc.) is observed, the survival function decreases by a factor corresponding to the proportion of individuals who experienced the event at that time.

We observe a sample (Y_i, D_i) for $i = 1, \dots, n$, where:

$$Y_i = \min(T_i, C_i), \quad D_i = \mathbb{1}_{\{T_i \leq C_i\}}.$$

We assume that the n individuals are independent and that censoring is **non-informative**: $T \perp\!\!\!\perp C$.

4.2 Construction of the Kaplan–Meier estimator

Definition 4.2.1 (Kaplan–Meier estimator). Let $t_{(1)} < t_{(2)} < \dots < t_{(m)}$ be the distinct times at which at least one event is observed ($D_i = 1$). We denote:

- d_j : the number of events observed at time $t_{(j)}$;
- n_j : the number of individuals still “at risk” just before $t_{(j)}$, that is, those such that $Y_i \geq t_{(j)}$.

The *Kaplan–Meier estimator* of the survival function is defined by:

$$\hat{S}_{\text{KM}}(t) = \prod_{j:t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right). \quad (4.1)$$

Each factor $\left(1 - \frac{d_j}{n_j}\right)$ represents the empirical probability of *surviving at time* $t_{(j)}$ given that one was still at risk just before that time. The product accumulates these probabilities over time, thus giving the probability of having experienced no event up to time t .

Graphically, $\hat{S}_{\text{KM}}(t)$ is a *decreasing step function*: it remains constant between two observed event times and drops by one step at each new event occurrence.

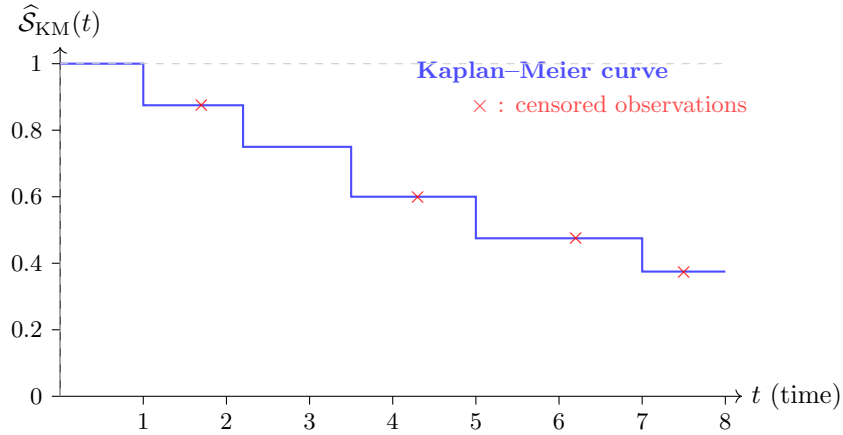


Figure 4.1: Example of a survival function estimated by the Kaplan–Meier approach with censoring indicators (red crosses).

The Kaplan–Meier expression can be read as a **cumulative product of conditional survival probabilities**. Indeed, for each time $t_{(j)}$ at which an event is observed:

$$\hat{S}_{\text{KM}}(t) = \prod_{j:t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right) = \underbrace{\left(1 - \frac{d_1}{n_1}\right)}_{\text{Survival up to } t_{(1)}} \times \underbrace{\left(1 - \frac{d_2}{n_2}\right)}_{\text{Survival from } t_{(1)} \text{ to } t_{(2)}} \times \dots \times \underbrace{\left(1 - \frac{d_j}{n_j}\right)}_{\text{Survival from } t_{(j-1)} \text{ to } t_{(j)}}.$$

Each factor $\left(1 - \frac{d_j}{n_j}\right)$ represents the empirical probability of surviving at $t_{(j)}$, conditionally on still being at risk just before that time. The product accumulates these elementary probabilities, reflecting the fact that total survival at time t is the probability of having “survived all previous stages”.

4.3 Fundamental properties

Propriété 4.3.1 (Nonparametric empirical estimator). *The Kaplan–Meier estimator is **non-parametric**: it does not assume any functional form for the survival distribution. It relies only on the observed proportions (d_j, n_j) and naturally incorporates right-censoring.*

Propriété 4.3.2 (Connection with likelihood). *The Kaplan–Meier estimator can be viewed as the nonparametric maximum likelihood estimator under the assumptions of independence and non-informative censoring. Indeed, maximizing the empirical likelihood of the observations (Y_i, D_i) leads to the product expression*

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

Proof idea. We show that the Kaplan–Meier expression maximizes the empirical likelihood under the assumptions:

$$T_1, \dots, T_n \text{ independent, } T_i \perp\!\!\!\perp C_i, \text{ and right-censoring.}$$

The full likelihood of the observations (Y_i, D_i) can be written as:

$$L = \prod_{i=1}^n [f(Y_i)]^{D_i} [\mathcal{S}(Y_i)]^{1-D_i},$$

where f and $\mathcal{S}$ denote the density and survival function of T .

Now group the observations according to the distinct event times $t_{(1)} < t_{(2)} < \dots < t_{(m)}$. Let d_j denote the number of events at $t_{(j)}$ and n_j the number of individuals still at risk just before $t_{(j)}$. Under a nonparametric approach, we assume that f (or the distribution of T) takes nonzero values only at the event times $t_{(j)}$: in other words, we seek to estimate directly the atomic probabilities

$$p_j = \mathbb{P}(T = t_{(j)} \mid T \geq t_{(j)}).$$

The empirical likelihood can then be rewritten as:

$$L = \prod_{j=1}^m p_j^{d_j} (1 - p_j)^{n_j - d_j}.$$

This expression corresponds exactly to a product of binomial distributions: at each time $t_{(j)}$, d_j successes (events) are observed among n_j individuals at risk.

Maximizing L with respect to p_j yields

$$\hat{p}_j = \frac{d_j}{n_j}.$$

Substituting this into the cumulative definition of survival gives:

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} (1 - \hat{p}_j) = \prod_{j: t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

We therefore recover the Kaplan–Meier formula as the *nonparametric maximum likelihood estimator* of the survival function. $\square$

The Kaplan–Meier estimator does not come out of nowhere: it follows directly from the likelihood principle, applied to an unknown and nonparametric distribution. The fundamental idea is the following: we seek to estimate the survival function $\mathcal{S}(t)$ without assuming any particular analytical form (exponential, Weibull, etc.). Instead of imposing a parametric density f_θ , we consider that the

distribution of T is *discrete* on the times at which events have been observed—in other words, all useful information is contained in the pairs (d_j, n_j) .

Thus, at each time $t_{(j)}$, we observe d_j events among n_j subjects still at risk. We can therefore view this situation as a small *binomial trial*:

“among n_j individuals at risk, d_j experience the event”.

The probability of an event at that time is denoted by p_j . By maximizing the empirical likelihood of these successive binomial observations, we naturally obtain

$$\hat{p}_j = \frac{d_j}{n_j}.$$

This result has a simple interpretation: it is the empirical proportion of events among those still exposed to risk at that precise moment.

Finally, the probability of *surviving up to time t* corresponds to the probability of having undergone *none of these successive small events*, hence the product:

$$\hat{S}(t) = \prod_{j:t_{(j)} \leq t} (1 - \hat{p}_j) = \prod_{j:t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

This multiplicative structure expresses exactly the cumulative probability of surviving at each stage of follow-up.

Thus, the Kaplan–Meier estimator appears both as:

- an **intuitive empirical construction**, based on observed proportions;
- and an **optimal solution in the likelihood sense**, obtained without imposing any parameter.

This dual status explains the simplicity, robustness, and popularity of this approach in survival analysis.

Propriété 4.3.3 (Extreme values and continuity). *The estimator satisfies:*

$$\mathcal{S}_{\text{KM}}(0) = 1, \quad \lim_{t \rightarrow +\infty} \mathcal{S}_{\text{KM}}(t) = \mathcal{S}_{\text{KM}}(t_{(m)}),$$

that is, it remains constant after the last observed event. It is right-continuous and decreasing, with discontinuities at event times.

Propriété 4.3.4 (Variance and confidence interval [Gre26]). *A classical estimate of the variance of $\mathcal{S}_{\text{KM}}(t)$ is given by Greenwood’s formula:*

$$\widehat{\text{Var}}[\mathcal{S}_{\text{KM}}(t)] = \mathcal{S}_{\text{KM}}(t)^2 \sum_{j:t_{(j)} \leq t} \frac{d_j}{n_j(n_j - d_j)}.$$

This variance can be used to construct a confidence interval for the survival function, for example via a logarithmic transformation:

$$\left[\log(-\log \hat{\mathcal{S}}_{\text{KM}}(t)) \pm z_{1-\alpha/2} \frac{\sqrt{\widehat{\text{Var}}[\mathcal{S}_{\text{KM}}(t)]}}{\hat{\mathcal{S}}_{\text{KM}}(t) \log \hat{\mathcal{S}}_{\text{KM}}(t)} \right].$$

with $z_{1-\alpha/2}$ the $(1 - \alpha/2)$ -quantile of a $\mathcal{N}(0, 1)$ distribution.

4.4 Interpretation and visualization

The Kaplan–Meier curve provides a simple and immediate representation of the empirical survival of a population. Each horizontal plateau corresponds to a period without an event, whereas each drop corresponds to one or more observed events. Censored observations often appear as marks (crosses, ticks) on the curve: they indicate subjects whose follow-up ended without the event occurring. This representation is particularly useful for:

- visually comparing several groups (e.g., treated vs untreated patients);
- estimating the median survival, defined by $\widehat{\text{Med}}(T) = \inf\{t : \widehat{S}(t) \leq 1/2\}$;
- quickly assessing the proportion of individuals still “alive” at a given time point;
- directly interpreting survival as a probability: for example, if $\widehat{S}(t) = 0.8$, this means that at time t , the estimated probability that an individual in the cohort is still alive (or has not yet experienced the event) is 80%.

4.5 Application: comparison between groups

Definition 4.5.1 (Group-specific survival function). Suppose that the population is divided into K distinct groups (e.g., treatments). For each group k , a survival function $S_k(t)$ is estimated using the Kaplan–Meier estimator computed from the observations of that group only:

$$\widehat{S}_k(t) = \prod_{j: t_{k,(j)} \leq t} \left(1 - \frac{d_{k,j}}{n_{k,j}}\right).$$

The log-rank test, introduced by Mantel [Man66] and formalized by the Peto brothers [PP72], makes it possible to compare two survival curves under the assumption of non-informative censoring.

Definition 4.5.2 (Log-rank test). We wish to test the hypothesis that two groups have the same survival function. The hypotheses are:

$$(H_0) : \quad \forall t \geq 0, S_1(t) = S_2(t),$$

$$(H_1) : \quad \exists t \geq 0, S_1(t) \neq S_2(t).$$

The *log-rank test* is based on comparing, at each event time $t_{(j)}$, the observed number of events in group 1 (d_{1j}) with the expected number e_{1j} under the null hypothesis H_0 . We then define the test statistic:

$$Z = \frac{\sum_j (d_{1j} - e_{1j})}{\sqrt{\sum_j v_{1j}}},$$

where v_{1j} is the variance associated with the difference $(d_{1j} - e_{1j})$ under H_0 .

Under the null hypothesis H_0 , the statistic

$$Z^2 \sim \chi_1^2 \quad (\text{asymptotically}).$$

We then deduce the ***p*-value**:

$$p\text{-value} = \mathbb{P}(\chi_1^2 \geq Z_{\text{obs}}^2),$$

which measures the probability of obtaining a statistic at least as extreme as the observed one, if H_0 were true.

The **decision rule** is as follows:

$$\text{if } p\text{-value} < \alpha, \quad \text{we reject } H_0,$$

at significance level α (often set at 5%). In that case, we conclude that there is a significant difference between the two survival curves.

Remark 7 (Practical interest). The log-rank test is the standard method for comparing survival curves. It is particularly robust and nonparametric: it does not assume any specific form of the hazard. However, it is mainly sensitive to global differences in survival and may lack power when the curves cross.

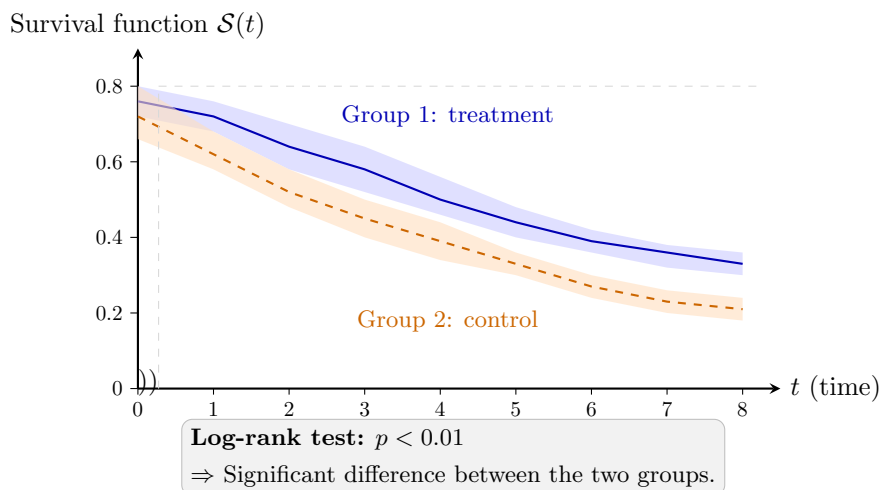


Figure 4.2: Illustrative example of a comparison between two estimated survival curves (Kaplan–Meier) with 95% confidence intervals. The result returned by the log-rank test makes it possible to conclude about the effectiveness of the treatment.

4.6 Summary and scope of the Kaplan–Meier approach

Summary. The Kaplan–Meier approach has several major advantages:

- it allows estimation of the survival function without any parametric assumption;
- it naturally incorporates right-censoring;
- it provides a simple, monotone, and interpretable estimate of $\mathcal{S}(t)$;
- it forms the basis for most between-group comparisons (log-rank test) and graphical representations of survival.

However, it does not allow one to account directly for continuous or categorical explanatory covariates: for that, one must turn to regression models, such as the Cox model, studied in the next chapter.

Chapter 5

Generalized linear regression

Survival analysis often relies on models in which the *relationship between covariates and the response* is not strictly linear. Before introducing the Cox model, it is useful to briefly recall the general structure of *generalized linear models* (GLMs), which unify linear, logistic, and Poisson regression approaches.

5.1 General form of the model

Definition 5.1.1 (Generalized linear model (GLM)). A *generalized linear model* relates the conditional mean of a response variable Y to a linear combination of covariates $\mathbb{X} = (X^1, \dots, X^p)$ through a link function g :

$$g(\mathbb{E}[Y \mid X^1, \dots, X^p]) = \beta_0 + \sum_{j=1}^p \beta_j X^j.$$

Here:

- $\mathbb{E}[Y \mid \mathbb{X}]$ denotes the mean of Y conditional on the covariates $\mathbb{X}$;
- g is a *link function* (e.g., identity, logit, log, etc.);
- $\beta_0, \dots, \beta_p$ are the model parameters (or *weights*) to be estimated.

The term “generalized linear” means that the *combination of covariates* remains linear, but the *relationship between this combination and the mean of Y* is modulated by a link function g . Thus, linearity holds in the space transformed by g , which makes it possible to handle continuous, binary, and discrete responses alike.

5.2 Examples of particular models

Example 5.2.1 (Classical linear regression). For a continuous response variable $Y \in \mathbb{R}$, one takes the identity link: $g(y) = y$, hence:

$$\mathbb{E}[Y \mid X] = \beta_0 + \sum_{j=1}^p \beta_j X^j.$$

This model assumes that the variance of Y is constant and independent of its mean.

Remark 8. Ordinary linear regression is therefore a particular case of the generalized linear model with $g(y) = y$. The term “generalized” adds nothing here, but it prepares the way for more complex links where the mean no longer depends linearly on the covariates.

Example 5.2.2 (Logistic regression). When Y is binary ($Y \in \{0, 1\}$), we seek to model the probability $p(x) = \mathbb{P}(Y = 1 \mid X = x)$. The logistic model uses the *logit* link:

$$g(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{j=1}^p \beta_j X^j.$$

We therefore obtain:

$$p(x) = \frac{\exp(\beta_0 + \beta^\top X)}{1 + \exp(\beta_0 + \beta^\top X)}.$$

Remark 9. The logit link makes linear regression compatible with a response bounded between 0 and 1. Each coefficient β_j measures the unit effect of X^j on the *log-odds* of success:

$$\log\left(\frac{p}{1-p}\right).$$

A one-unit variation in X^j therefore multiplies the *odds* by $\exp(\beta_j)$.

Example 5.2.3 (Poisson regression). To model counts ($Y \in \mathbb{N}$), one assumes:

$$Y \mid X \sim \text{Poisson}(h(X)), \quad \text{with } g(h) = \log(h).$$

Thus:

$$\log \mathbb{E}[Y \mid X] = \beta_0 + \sum_{j=1}^p \beta_j X^j.$$

Remark 10. Poisson regression is conceptually very close to the models used in survival analysis: it relates an intensity (here, $h(X)$) to a linear predictor through a logarithmic transformation. This analogy naturally prepares the ground for the Cox model, in which the *instantaneous hazard function* will play a role analogous to $h(X)$.

5.3 Toward the Cox model

Propriété 5.3.1 (Principle shared by GLMs and the Cox model). *In a GLM as in the Cox model, one seeks to relate a quantity of interest (mean, risk, intensity) to covariates through a linear predictor:*

$$\text{Quantity of interest} = g^{-1}(\beta_0 + \beta^\top X).$$

The main difference is that, in the Cox model, the quantity under study is no longer a mean or a probability, but a time-dependent instantaneous hazard rate.

Generalized linear models therefore provide the conceptual framework within which the Cox model is situated: the linearity of the effects of covariates is preserved, but the target function becomes the *logarithm of the instantaneous hazard*. It is this idea—a semiparametric model with a multiplicative link on the hazard—that we will develop in the next chapter.

Chapter 6

The Cox model: foundations and estimation

The Cox model [Cox72], proposed in 1972, is one of the cornerstones of modern survival analysis. It provides a flexible framework for studying the effect of covariates on the risk of an event, without assuming any particular form for the distribution of survival times.

This model relies on a so-called *separability* assumption: the dependence on time and the dependence on covariates are multiplicative in the hazard function. Before introducing this idea, let us first recall what the term *semiparametric* means.

6.1 Semiparametric models

Definition 6.1.1 (Semiparametric model). A *semiparametric model* is a model that contains both:

- a **parametric** component, of finite dimension (here, the coefficients $\beta \in \mathbb{R}^p$);
- a **nonparametric** component, of infinite dimension (here, the baseline hazard function $h_0(t)$).

Thus, unlike a fully parametric model (e.g., exponential, Weibull), a semiparametric model does not assume any particular form for the distribution of the survival time.

In a semiparametric model, one seeks to capture the systematic effects of covariates through a parametric structure while leaving the temporal dynamics unspecified. This compromise combines *interpretability* and *flexibility*, especially for the modeling of duration phenomena where the form of the survival function is often unknown.

6.2 Separability model

Definition 6.2.1 (Cox model [Cox72]). The Cox model postulates a **multiplicative separability** of the instantaneous hazard:

$$h(t \mid \mathbb{X}) = h_0(t) \exp(\beta^\top \mathbb{X}), \quad (6.1)$$

where:

- $h_0(t)$ is the **baseline hazard function** (nonparametric), describing the temporal dynamics common to all individuals;

- $\exp(\beta^\top \mathbb{X})$ is a **multiplicative factor** describing the effect of covariates on the relative hazard;
- the linear form $\beta^\top \mathbb{X}$, often denoted by η , is called the **risk score**.

This “separability” assumption is analogous to what is encountered in physics: the hazard depends both on a *global temporal factor* $h_0(t)$ and on a *structural factor* specific to each individual. In other words, the time dynamics (the “danger profile” of the system) are separated from the effect of the observed characteristics. This multiplicative decomposition makes the model particularly interpretable.

Propriété 6.2.1 (Relative hazard). *For two individuals i and j with covariates $\mathbb{X}_i$ and $\mathbb{X}_j$, the ratio of their instantaneous hazards is:*

$$\frac{h(t \mid \mathbb{X}_i)}{h(t \mid \mathbb{X}_j)} = \exp(\beta^\top (\mathbb{X}_i - \mathbb{X}_j)),$$

independently of time t .

This property illustrates the fundamental assumption of the Cox model: the relative hazards between individuals are constant over time. The survival curves may therefore differ vertically (difference in level) but not horizontally (difference in shape).

6.3 Partial likelihood

Definition 6.3.1 (Cox partial likelihood). Rather than modeling $h_0(t)$, Cox introduced the *partial likelihood* in 1975 [Cox75], which makes it possible to estimate β without specifying $h_0(t)$. Let $t_{(1)} < \dots < t_{(m)}$ be the distinct observed event times. The partial likelihood is written as:

$$L_p(\beta) = \prod_{j=1}^m \frac{\exp(\beta^\top \mathbb{X}_j)}{\sum_{i \in \mathcal{R}(t_{(j)})} \exp(\beta^\top \mathbb{X}_i)}, \quad (6.2)$$

where $\mathcal{R}(t_{(j)})$ denotes the set of individuals still at risk just before $t_{(j)}$.

The idea is to compare, at each event time, the probability that the individual i who fails is the one experiencing the event among all those still at risk. The baseline hazard function $h_0(t)$ disappears in the ratio, which makes it possible to estimate β without any parametric assumption on it. Moreover, the form of the Cox partial likelihood strongly resembles the *softmax* function used in statistical learning. Indeed, expression (6.2) corresponds exactly to an exponential normalization of the scores $\beta^\top \mathbb{X}_j$, that is, to a *softmax* computed over the set of individuals at risk.

Thus, the Cox partial likelihood can be interpreted as a sum of *softmax*-type log-likelihoods, where each at-risk individual receives a score $\exp(\beta^\top \mathbb{X}_i)$, and the probability of being the observed event at a given time corresponds to the normalization of these scores among all individuals still in competition.

Propriété 6.3.1 (Partial log-likelihood). *In practice, one maximizes the logarithm of $L_p(\beta)$:*

$$\ell_p(\beta) = \sum_{j=1}^m \left[\beta^\top \mathbb{X}_{(j)} - \log \left(\sum_{i \in \mathcal{R}(t_{(j)})} \exp(\beta^\top \mathbb{X}_i) \right) \right]. \quad (6.3)$$

Remark 11 (Connection with the full likelihood). The partial likelihood is derived from the full likelihood of the Cox model after simplification of the terms common to all individuals. It retains only the components that depend on β , and therefore on the effects of the covariates.

Propriété 6.3.2 (Estimation of the coefficient vector $\hat{\beta}$). *The estimator of the coefficient vector is obtained by maximum likelihood as expressed in (6.3):*

$$\hat{\beta} = \operatorname{argmax}_{\beta \in \mathbb{R}^p} \ell_p(\beta),$$

that is, the point that maximizes the likelihood (or minimizes the negative log-likelihood).

Remark 12 (Maximum condition and interpretation). The optimum $\hat{\beta}$ satisfies the first-order condition:

$$\left. \frac{\partial \ell_p(\beta)}{\partial \beta} \right|_{\beta=\hat{\beta}} = 0,$$

that is, componentwise:

$$\sum_{j=1}^m \left[\mathbb{X}_{(j)} - \frac{\sum_{i \in \mathcal{R}(t_{(j)})} \mathbb{X}_i \exp(\hat{\beta}^\top \mathbb{X}_i)}{\sum_{i \in \mathcal{R}(t_{(j)})} \exp(\hat{\beta}^\top \mathbb{X}_i)} \right] = 0.$$

This equation expresses a balance between the covariates observed for the subjects who experienced the event and their weighted average among all individuals still at risk at the same time.

Intuitively, the model seeks a vector $\hat{\beta}$ such that, at each event time, individuals with a high risk score $\beta^\top \mathbb{X}$ have a higher probability of being the observed event. In other words, $\hat{\beta}$ adjusts the boundary between “low risk” and “high risk” so as to maximize the likelihood of the observed event ordering.

Remark 13 (Numerical optimization). In practice, the maximum of $\ell_p(\beta)$ is obtained numerically by Newton–Raphson-type algorithms [TG00]:

$$\beta^{(k+1)} = \beta^{(k)} + \mathcal{I}(\beta^{(k)})^{-1} U(\beta^{(k)}),$$

where $U(\beta)$ is the score vector (gradient of the partial log-likelihood) and $\mathcal{I}(\beta)$ is the observed information matrix (see (8.2)). This iterative scheme converges toward $\hat{\beta}$ under standard regularity conditions.

6.4 Estimation of the baseline hazard function (Breslow method)

Definition 6.4.1 (Breslow estimator [Bre74]). Once $\hat{\beta}$ has been estimated, the baseline cumulative hazard function $H_0(t)$ can be estimated by the Breslow method:

$$\hat{H}_0(t) = \sum_{j: t_{(j)} \leq t} \frac{d_j}{\sum_{i \in \mathcal{R}(t_{(j)})} \exp(\hat{\beta}^\top \mathbb{X}_i)},$$

where d_j is the number of events observed at time $t_{(j)}$, i.e., $d_j = \sum_{i=1}^n \mathbb{1}_{\{T_i=t_{(j)}, D_i=1\}}$.

Each numerator term (d_j) corresponds to the number of events actually observed at time $t_{(j)}$. The denominator represents the *total exposed risk* at that time, weighted by the estimated effects of the covariates. The fraction

$$\frac{d_j}{\sum_{i \in \mathcal{R}(t_{(j)})} \exp(\hat{\beta}^\top \mathbb{X}_i)}$$

can be interpreted as the empirical contribution to the baseline cumulative hazard.

Propriété 6.4.1 (Estimated survival function). *One then deduces the estimate of the survival function for an individual i with covariates $\mathbb{X}_i$:*

$$\hat{S}(t \mid \mathbb{X}_i) = \exp \left[-\hat{H}_0(t) \exp \left(\hat{\beta}^\top \mathbb{X}_i \right) \right].$$

This expression illustrates the power of the Cox model:

- the nonparametric part $\hat{H}_0(t)$ captures the temporal dynamics;
- the parametric part $\exp \left(\hat{\beta}^\top \mathbb{X} \right)$ captures the effect of the covariates;
- their product expresses the *separability* of the hazard.

It is this duality, combining temporal flexibility and covariate linearity, that explains the longevity and popularity of the Cox model.

Chapter 7

Penalized Cox models

Classical Cox models perform well when the number of covariates p remains moderate compared with the number of observations n . However, in many modern settings (genomic data, imaging, machine learning), one often has *many more explanatory variables than observed subjects*. In this case, maximizing the classical partial likelihood becomes unstable, or even impossible.

One solution consists in introducing a **penalty term** on the coefficients β , in order to control model complexity. This is the principle of penalized Cox models.

7.1 General principle

Definition 7.1.1 (Penalized partial log-likelihood). The *penalized partial log-likelihood* is defined as:

$$\ell_p^{(\text{pen})}(\beta) = \ell_p(\beta) - \lambda P(\beta),$$

where:

- $\ell_p(\beta)$ is the Cox partial log-likelihood,
- $P(\beta)$ is a penalty term measuring the complexity of the vector β ,
- $\lambda > 0$ is a regularization parameter controlling the strength of the penalty.

Penalization acts as a constraint: it limits the magnitude of the coefficients to avoid overfitting and improve numerical stability. The parameter λ controls the trade-off between *bias* and *variance*:

- $\lambda = 0$: classical Cox model;
- large λ : smoother model, smaller coefficients, possibly equal to zero.

7.2 Common types of penalties

LASSO penalty (ℓ_1)

Definition 7.2.1 (Cox-LASSO). The LASSO-type penalty (Least Absolute Shrinkage and Selection Operator), introduced by Robert Tibshirani [Tib97], consists in adding the ℓ_1 norm of the coefficients:

$$P_{\text{LASSO}}(\beta) = \|\beta\|_1 = \sum_{j=1}^p |\beta_j|.$$

The penalized partial log-likelihood becomes:

$$\ell_p^{(\text{LASSO})}(\beta) = \ell_p(\beta) - \lambda \sum_{j=1}^p |\beta_j|.$$

Remark 14 (Interest of the LASSO). This penalty promotes **sparsity**: some coefficients are shrunk exactly to zero. It is particularly useful when $p \gg n$ (a high-dimensional setting where the number of parameters to estimate exceeds the number of observations), since it simultaneously performs:

- automatic *variable selection*;
- *regularization* of the solution to avoid overfitting.

The LASSO is therefore well suited for high-dimensional data, where only a small proportion of co-variates is truly informative.

Ridge penalty (ℓ_2)

Definition 7.2.2 (Cox-ridge [VvH94]). The ridge penalty relies on the quadratic norm:

$$P_{\text{ridge}}(\beta) = \|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2.$$

The model then becomes:

$$\ell_p^{(\text{ridge})}(\beta) = \ell_p(\beta) - \lambda \sum_{j=1}^p \beta_j^2.$$

Remark 15 (Interest of ridge). This penalty does not set coefficients exactly to zero, but shrinks them continuously. It is effective when several covariates are correlated, since it *stabilizes the estimates* by distributing the weight across collinear variables. Ridge therefore favors *robustness* rather than sparsity.

Elastic Net penalty

Definition 7.2.3 (Cox-Elastic Net [FHT10]). The *Elastic Net* penalty combines the effects of LASSO and ridge:

$$P_{\text{EN}}(\beta) = \alpha \|\beta\|_1 + \frac{(1-\alpha)}{2} \|\beta\|_2^2,$$

where $0 \leq \alpha \leq 1$ controls the relative contribution of the two terms. The penalized partial log-likelihood then becomes:

$$\ell_p^{(\text{EN})}(\beta) = \ell_p(\beta) - \lambda \left[\alpha \sum_{j=1}^p |\beta_j| + \frac{(1-\alpha)}{2} \sum_{j=1}^p \beta_j^2 \right].$$

Remark 16 (Interest of the Elastic Net). This compromise inherits:

- the **sparsity** of LASSO (selection of relevant variables);
- the **stability** of ridge (handling correlated covariates).

It is the preferred model when explanatory variables exhibit strong correlations or when one seeks a balance between selection and robustness.

Remark on the choice of λ . The regularization parameter λ is generally selected by *cross-validation*, by maximizing the average partial log-likelihood on validation samples. An optimal value of λ helps minimize overfitting while retaining the most informative variables.

Chapter 8

Inference and interpretation in the Cox model

This section is devoted to the quantification of the uncertainty associated with the estimated coefficients $\hat{\beta}$ of the Cox model, as well as to their statistical and practical interpretation. Although the Cox model is semi-parametric, it allows classical inference through the partial log-likelihood and the observed information matrix.

8.1 Confidence intervals and estimation uncertainty

Definition 8.1.1 (Estimator and asymptotic variance). Under standard regularity conditions and the proportional hazards assumption, the partial maximum likelihood estimator $\hat{\beta}$ satisfies:

$$\hat{\beta} \xrightarrow{\text{asympt.}} \mathcal{N}(\beta, \mathcal{I}(\beta)^{-1}), \quad (8.1)$$

where $\mathcal{I}(\beta)$ is the *Fisher information matrix*. In practice, it is estimated by the observed information matrix:

$$\hat{\mathcal{I}}(\hat{\beta}) = - \frac{\partial^2 \ell_p(\beta)}{\partial \beta \partial \beta^\top} \Big|_{\beta=\hat{\beta}}. \quad (8.2)$$

Consequently, the covariance matrix estimator is:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\mathcal{I}}(\hat{\beta})^{-1}, \quad \widehat{\text{Var}}(\hat{\beta}_j) = [\hat{\mathcal{I}}(\hat{\beta})^{-1}]_{jj}. \quad (8.3)$$

This Gaussian approximation follows from likelihood theory: as the sample size increases, the distribution of $\hat{\beta}$ concentrates around the true value β . Thus, the uncertainty of each coefficient can be quantified from its estimated variance $\widehat{\text{Var}}(\hat{\beta}_j)$.

Remark 17 (Intuition on Fisher information). The Fisher information matrix measures the *amount of information* that the data provide about the parameters to be estimated. It is constructed from the second derivative (the Hessian) of the log-likelihood, which describes the *curvature* of the likelihood function near its maximum.

Intuitively:

- if the log-likelihood is sharply curved around its maximum, the peak is narrow: the estimator $\hat{\beta}$ is precise and its variance is small;
- if the log-likelihood is flat, small variations of β produce little change in likelihood: the uncertainty on $\hat{\beta}$ is large.

Thus, the second derivative $-\frac{\partial^2 \ell_p}{\partial \beta^2}$ quantifies how rapidly the likelihood decreases away from its maximum. The stronger this curvature, the more precise the estimation.

Propriété 8.1.1 (Asymptotic confidence interval). *For each coefficient β_j , an asymptotic confidence interval at level $1 - \alpha$ is:*

$$\widehat{\beta}_j \pm z_{1-\alpha/2} \sqrt{\widehat{\text{Var}}(\widehat{\beta}_j)},$$

where $z_{1-\alpha/2}$ is the quantile of the standard normal distribution $\mathcal{N}(0, 1)$ (e.g., $z_{0.975} \approx 1.96$ for a 95% confidence level).

These intervals quantify the uncertainty associated with the estimated effect of a covariate on the hazard function. If the confidence interval includes 0, one cannot reject the hypothesis that the covariate has no significant effect on the hazard.

8.2 Significance testing and interpretation of coefficients

Wald test

Definition 8.2.1 (Wald test). The Wald test evaluates, for each coefficient β_j , the hypothesis:

$$\begin{cases} H_0 : \beta_j = 0, \\ H_1 : \beta_j \neq 0. \end{cases}$$

The test statistic is:

$$Z_j = \frac{\widehat{\beta}_j}{\sqrt{\widehat{\text{Var}}(\widehat{\beta}_j)}}, \quad Z_j^2 \sim \chi_1^2 \text{ under } H_0.$$

The Wald test evaluates whether the ratio “estimate / uncertainty” is sufficiently large to conclude that the covariate has a significant effect on the hazard.

Likelihood ratio test

Definition 8.2.2 (Likelihood ratio test). We compare the partial log-likelihood of the full model $\ell_p(\widehat{\beta})$ with that of the reduced model without the tested covariate ($\ell_p(\beta_j = 0)$). The test statistic is:

$$\Lambda = 2[\ell_p(\widehat{\beta}) - \ell_p(\beta_j = 0)], \quad \Lambda \sim \chi_1^2 \text{ under } H_0.$$

Remark 18 (Interpretation). This test measures the loss of fit induced by constraining $\beta_j = 0$. A large decrease in likelihood indicates that the covariate significantly improves predictive performance.

Score test (Rao test)

Definition 8.2.3 (Score test). The score test relies on the gradient of the partial log-likelihood evaluated under H_0 . The statistic is:

$$U(\beta_0) = \left. \frac{\partial \ell_p(\beta)}{\partial \beta} \right|_{\beta=\beta_0}, \quad S = U(\beta_0)^\top \mathcal{I}(\beta_0)^{-1} U(\beta_0),$$

with

$$S \sim \chi_q^2 \text{ under } H_0,$$

where q is the number of tested coefficients.

Comparison of the three tests. The Wald, likelihood ratio, and score tests are asymptotically equivalent. In practice:

- the Wald test is simple but can be unstable when coefficients are large or highly correlated;
- the likelihood ratio test is generally more robust and often preferred in software implementations;
- the score test is useful when the full model has not yet been fitted (local evaluation near H_0).

8.3 Interpretation of coefficients and hazard ratio

Definition 8.3.1 (Hazard ratio). In the Cox model (6.1), the hazard ratio associated with a one-unit increase in covariate X_j is:

$$\text{HR}_j = \exp(\beta_j).$$

Remark 19 (Interpretation of the hazard ratio). A simply way to interpret the Hazard Ratio:

- If $\beta_j > 0$, then $\text{HR}_j > 1$: the covariate increases the instantaneous hazard;
- If $\beta_j < 0$, then $\text{HR}_j < 1$: the covariate decreases the instantaneous hazard;
- If $\beta_j = 0$, then $\text{HR}_j = 1$: the covariate has no effect.

Thus, the hazard ratio measures the multiplicative effect of a covariate on the hazard at all times, independently of time (proportional hazards assumption).

Propriété 8.3.1 (Confidence interval for the hazard ratio). *The $100(1-\alpha)\%$ confidence interval for HR_j is obtained directly from that of β_j :*

$$\left[\exp \left(\hat{\beta}_j - z_{1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\beta}_j)} \right), \exp \left(\hat{\beta}_j + z_{1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\beta}_j)} \right) \right].$$

This interval quantifies the uncertainty on the relative effect of a covariate on the hazard. For example, $\text{HR} = 1.5$ indicates a 50% increase in instantaneous risk; however, if the confidence interval includes 1, this increase is not statistically significant.

Chapter 9

Evaluation and performance metrics in survival analysis

Evaluating a survival model is an essential step in any applied study. As in standard regression or classification models, it is necessary to measure the goodness of fit, predictive ability, and reliability of the resulting estimates. However, the presence of censored data makes this evaluation more complex: not all observations contribute in the same way to the likelihood, nor to the performance metrics.

This chapter presents the main measures used to quantify the performance of a survival model, with an emphasis on two fundamental types of metrics:

1. **discrimination**: the model's ability to correctly rank individuals according to their risk;
2. **calibration**: the agreement between predicted probabilities and the observed proportions of events.

We conclude with a reminder on goodness-of-fit criteria (AIC, BIC) and a comparison with the Kaplan–Meier model.

9.1 Motivations and challenges of performance metrics

Definition 9.1.1 (Purpose of performance metrics). A *performance metric* evaluates the predictive quality of a model according to a specific criterion (discrimination, calibration, likelihood, etc.). It allows one to compare several models, tune hyperparameters, and assess the robustness of an estimator.

A good survival model must not only fit the training data properly: it must also be able to *predict correctly the ordering of future events*, and to *provide probabilities consistent with the observed reality*. This is precisely what discrimination and calibration indices measure.

Remark 20 (Limitations and precautions). No single metric is universally sufficient. A model may discriminate patients well (high C-index) but be poorly calibrated (high Brier Score). It is therefore recommended to assess both aspects simultaneously in order to judge the overall performance of the model.

9.2 Discrimination

9.2.1 The concordance index (*C-index*)

Definition 9.2.1 (C-index). The concordance index measures the model's ability to correctly rank subjects according to their predicted risk. Formally:

$$\text{C-index} = \frac{\sum_{i,j} \mathbb{1}_{\{T_j < T_i\}} \cdot \mathbb{1}_{\{\hat{\eta}_j > \hat{\eta}_i\}} \cdot \delta_j}{\sum_{i,j} \mathbb{1}_{\{T_j < T_i\}} \cdot \delta_j},$$

where T_i and T_j are the observed times, δ_j is the event indicator, and $\hat{\eta}_i$ is the estimated risk score for individual i .

Remark 21 (Interpretation). The C-index represents the proportion of pairs of individuals for which the model correctly ranks the subject who is at earlier risk as having a higher risk score. A value of:

- 0.5 indicates random performance;
- 1 corresponds to perfect discrimination.

Remark 22 (Connection with the AUC). In the absence of censoring, the C-index coincides with the area under the ROC curve (AUC). It is therefore a natural generalization of the AUC to the survival setting, adapted to partial comparisons.

9.2.2 Time-dependent AUC

Definition 9.2.2 (Time-dependent AUC). To evaluate discrimination at a given time t , one defines a time-dependent AUC:

$$\widehat{\text{AUC}}(t) = \frac{\sum_{i,j} \mathbb{1}_{\{T_i > t\}} \mathbb{1}_{\{T_j \leq t\}} \cdot \mathbb{1}_{\{\hat{\eta}_j > \hat{\eta}_i\}} \delta_j(t)}{\sum_{i,j} \mathbb{1}_{\{T_i > t\}} \mathbb{1}_{\{T_j \leq t\}} \cdot \delta_j(t)}.$$

Remark 23 (Intuition). This metric evaluates the model's ability, at a given time t , to distinguish subjects who are still alive ($T_i > t$) from those who experienced the event before t . It provides a time-specific measure of discrimination, often represented as a curve $t \mapsto \widehat{\text{AUC}}(t)$.

9.3 Calibration

9.3.1 Brier Score

Definition 9.3.1 (Censored Brier Score). The Brier Score quantifies the accuracy of the predicted survival probability at horizon t :

$$\text{BS}(t) = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \left[\mathbb{1}_{\{T_i \leq t\}} \frac{(0 - \hat{S}(t | \mathbb{X}_i))^2}{\hat{G}(T_i)} \delta_i + \mathbb{1}_{\{T_i > t\}} \frac{(1 - \hat{S}(t | \mathbb{X}_i))^2}{\hat{G}(t)} \right],$$

where $\hat{G}$ denotes the censoring survival function estimated by Kaplan–Meier.

Remark 24 (Interpretation). The Brier Score measures the quadratic discrepancy between predicted survival and observed survival. It takes values in $[0, 1]$, and the smaller it is, the better the calibration. A perfectly calibrated model would have $\text{BS}(t) = 0$ for all t .

9.3.2 Calibration plot

Definition 9.3.2 (Calibration plot). The *calibration plot* compares, for a horizon t^* , the predicted survival probabilities $\hat{S}(t^* | \mathbb{X}_i)$ with the observed proportions of individuals actually alive at that time. One typically plots:

Predicted probability vs. Observed probability.

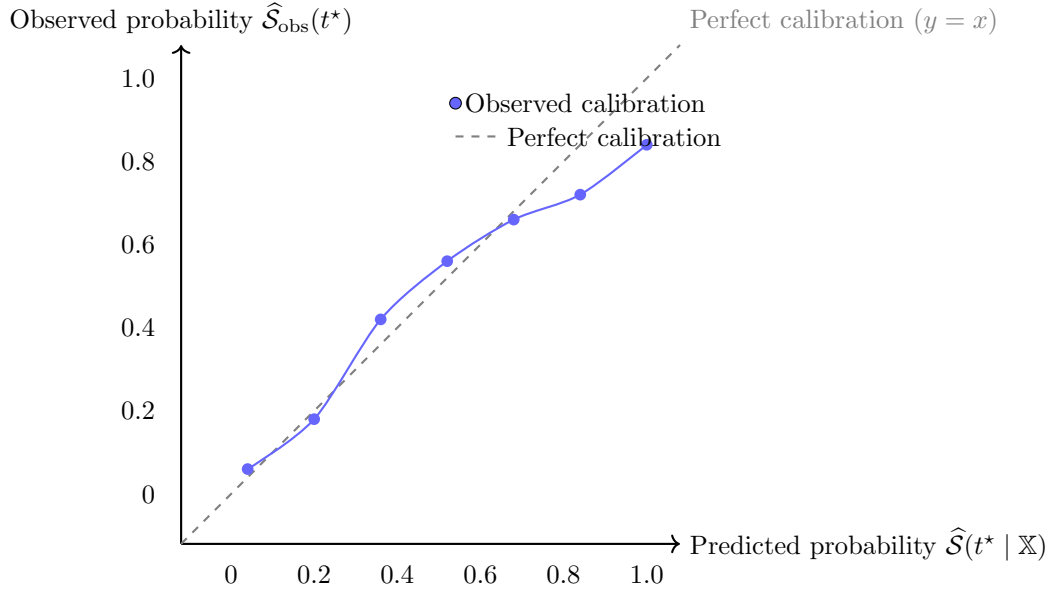


Figure 9.1: Example of a *calibration plot*: the points represent the observed survival by patient group, and the gray line indicates perfect calibration.

Remark 25 (How to read the plot). A curve close to the diagonal $y = x$ indicates good calibration. A curve above the diagonal indicates an *overestimation of risk* (the model is too pessimistic), whereas a curve below the diagonal indicates an *underestimation of risk*.

9.4 Information criteria and overall goodness of fit

Definition 9.4.1 (AIC and BIC criteria). To compare several models fitted on the same cohort, one often uses:

$$\text{AIC} = -2\ell_p(\hat{\beta}) + 2p, \quad \text{BIC} = -2\ell_p(\hat{\beta}) + p \log(n),$$

where p is the number of estimated parameters and n the number of observations.

Remark 26 (Interpretation). These criteria penalize model complexity: the more parameters a model has, the larger the penalty. One chooses the model with the smallest AIC or BIC value, reflecting an optimal trade-off between goodness of fit and parsimony.

Chapter 10

Application and interpretation of a survival model

Once a survival model has been estimated — whether it is a nonparametric Kaplan–Meier model $\hat{S}_{\text{KM}}(t)$ or a semiparametric Cox model

$$\hat{h}(t \mid \mathbb{X}) = \hat{h}_0(t) \exp(\mathbb{X}^\top \hat{\boldsymbol{\beta}}),$$

— it becomes essential to interpret, validate, and exploit the results for clinical decision-making, cohort follow-up, or the understanding of risk factors.

10.1 Overall survival of a cohort

Definition 10.1.1 (Marginal survival of the cohort). The *estimated overall survival* of a cohort corresponds to the average of the individual survival probabilities:

$$\hat{S}_{\text{Cox}}(t) = \frac{1}{n} \sum_{i=1}^n \hat{S}(t \mid \mathbb{X}_i) = \frac{1}{n} \sum_{i=1}^n \exp\left(- \int_0^t \hat{h}_0(u) \exp(\mathbb{X}_i^\top \hat{\boldsymbol{\beta}}) du\right).$$

Remark 27. This quantity is comparable to the Kaplan–Meier curve, which provides a nonparametric empirical estimate of survival in the population. Comparing $\hat{S}_{\text{Cox}}(t)$ with $\hat{S}_{\text{KM}}(t)$ makes it possible to assess the overall calibration of the model.

10.2 Risk stratification

Definition 10.2.1 (Risk score and stratification). In the Cox model, each individual i is assigned a *linear risk score*:

$$\eta_i = \mathbb{X}_i^\top \hat{\boldsymbol{\beta}}.$$

This score reflects the individual’s relative tendency to experience the event, all else being equal. Risk stratification consists in partitioning the cohort according to quantiles of the score η_i (e.g., low, intermediate, high).

Remark 28. This approach makes it possible to visualize distinct survival curves by risk group, and to assess the model’s ability to discriminate between prognostic profiles. In practice, risk stratification is used to define differentiated therapeutic strategies: intensifying follow-up or adjusting treatment for high-risk groups.

10.3 Interpretation of the coefficients

Propriété 10.3.1 (Multiplicative interpretation). *In the Cox model, the coefficients $\hat{\beta}_j$ are interpreted through the hazard ratios:*

$$HR_j = \exp(\hat{\beta}_j).$$

An $HR_j > 1$ means that variable X_j increases the instantaneous risk of the event (unfavorable factor), whereas an $HR_j < 1$ indicates a protective effect.

Remark 29. The interpretation remains *conditional on the model*, that is, on the proportional hazards assumption. Confidence intervals for $\hat{\beta}_j$ and significance tests (Wald, score, or likelihood ratio) make it possible to assess the robustness of these effects.

10.4 Individual survival prediction

Definition 10.4.1 (Predicted individual survival). For an individual with characteristics $\mathbb{X}^*$, the predicted survival function is:

$$\hat{S}(t \mid \mathbb{X}^*) = \exp\left(-\int_0^t \hat{h}_0(u) \exp(\mathbb{X}^{*\top} \hat{\beta}) du\right).$$

Remark 30. In practice, individual survival prediction is subject to substantial uncertainty, because it depends both on the variance of the coefficients and on the nonparametric estimation of $\hat{h}_0(t)$. A relative interpretation (comparison of two profiles) is therefore often preferred over an absolute survival probability.

10.5 Extension: mixed-effects models and within-subject dependence

Definition 10.5.1 (Cox model with random effects). For longitudinal or hierarchical data (several observations per patient), one may introduce a random term b_i :

$$h(t \mid \mathbb{X}_i, b_i) = h_0(t) \exp(\mathbb{X}_i^\top \beta + b_i), \quad b_i \sim \mathcal{N}(0, \sigma_b^2).$$

Remark 31. This type of model, called a *mixed-effects model*, makes it possible to capture inter-individual heterogeneity and within-subject correlation. It lies at the interface between survival models and Bayesian hierarchical models. Such models require more complex estimation procedures (via the EM algorithm or MCMC), but they offer a more realistic description of patient trajectories.

Chapter summary

- The Kaplan–Meier curve or the marginal survival from the Cox model describes the overall survival of the cohort.
- The linear score $\eta_i = \mathbb{X}_i^\top \hat{\beta}$ allows risk stratification and supports clinical decision-making.
- The coefficients are interpreted in terms of hazard ratios, which measure the relative impact of the covariates.
- Individual predictions are possible but uncertain, especially in the long term.
- Mixed-effects extensions provide advanced modeling for longitudinal data.

Chapter 11

Generalized hazard models: beyond the linear model

The Cox model relies on a fundamental assumption of *linearity* of the risk predictor:

$$\eta(\mathbb{X}) = \mathbb{X}^\top \beta.$$

Although practical and interpretable, this assumption may be too restrictive when the relationship between the covariates and the risk is not linear. It is therefore natural to consider extensions in which the hazard function depends on a nonlinear predictor:

$$\eta(\mathbb{X}) = g(\mathbb{X}),$$

while retaining the structure of the partial likelihood.

11.1 General principle

Definition 11.1.1 (Generalized survival model). A generalized proportional hazards survival model is defined by:

$$h(t \mid \mathbb{X}) = h_0(t) \exp(g(\mathbb{X})),$$

where $g : \mathbb{R}^p \rightarrow \mathbb{R}$ is a function (possibly nonlinear) that learns to estimate the *risk score* from the covariates.

Remark 32. The role of $g(\mathbb{X})$ is analogous to that of the linear predictor $\mathbb{X}^\top \beta$ in the classical Cox model. The function g may be parametric (logistic regression, spline, MLP, etc.) or nonparametric (random forest, gradient boosting).

11.2 Generalized partial likelihood

Definition 11.2.1 (Generalized partial likelihood). The Cox principle remains valid: at each time $t_{(j)}$, the probability that individual i is the one experiencing the event is given by:

$$\mathbb{P}(i \text{ fails at } t_{(j)}) = \frac{\exp(g(\mathbb{X}_i))}{\sum_{k \in \mathcal{R}(t_{(j)})} \exp(g(\mathbb{X}_k))}.$$

The *generalized partial likelihood* is therefore written as:

$$L_p(g) = \prod_{j=1}^m \frac{\exp(g(\mathbb{X}_{(j)}))}{\sum_{i \in \mathcal{R}(t_{(j)})} \exp(g(\mathbb{X}_i))}.$$

Remark 33. One observes that $h_0(t)$ still disappears in the ratio: it is this property of *hazard separability* that makes the Cox model robust to the functional form of the baseline hazard.

11.3 Optimization step

Definition 11.3.1 (Maximization of the partial log-likelihood). The estimation of g relies on maximizing the partial log-likelihood:

$$\ell_p(g) = \sum_{j=1}^m \left[g(\mathbb{X}_{(j)}) - \log \left(\sum_{i \in \mathcal{R}(t_{(j)})} \exp(g(\mathbb{X}_i)) \right) \right].$$

Remark 34. Optimizing $\ell_p(g)$ does not require explicit knowledge of $h_0(t)$. It is therefore naturally suited to numerical optimization by gradient methods (for neural networks) or by functional iterations (for forests or boosting).

11.4 Examples of functions $g(\mathbb{X})$

11.4.1 Classical linear case

$$g(\mathbb{X}) = \mathbb{X}^\top \beta.$$

This is the standard Cox model. The optimization is convex and yields a solution numerically via Newton–Raphson.

11.4.2 Random forest case

$$g(\mathbb{X}) = \log \left(\frac{1}{T} \sum_{t=1}^T \exp(f_t(\mathbb{X})) \right),$$

where each f_t is a decision tree. This type of model corresponds to *Random Survival Forests* [IKBL08], where the risk is estimated by averaging the tree-based submodels.

11.4.3 Neural network case

$$g(\mathbb{X}) = \text{NN}_\theta(\mathbb{X}),$$

where NN_θ denotes a neural network fully parameterized by θ . One maximizes $\ell_p(g_\theta)$ by gradient descent, which defines the *DeepSurv* model [KSC⁺18].

Remark 35. The term $\text{NN}_\theta(\mathbb{X})$ plays the same role as $\mathbb{X}^\top \beta$: it provides a differentiable risk score optimized through the partial likelihood.

11.4.4 Parametric case

One may specify a known analytical form, for example:

$$g(\mathbb{X}) = \sum_{j=1}^p \beta_j \sin(\omega_j X_j),$$

or any other parametric function adapted to the physical or biological problem under consideration. This type of modeling preserves flexibility while maintaining interpretability.

11.5 Advantages and practical considerations

- **Flexibility:** these extensions make it possible to capture complex relationships between covariates and risk.
- **Invariant principle:** maximizing the partial likelihood remains the core of the estimation process.
- **Comparability:** the scores $g(\mathbb{X})$, although nonlinear, retain a monotone interpretation of relative risk.
- **Calibration and validation:** the same tools (C-index, Brier score, calibration curves) apply for performance evaluation.

Chapter summary

- The Cox model can be generalized by replacing the linear predictor $\mathbb{X}^\top \beta$ with a nonlinear function $g(\mathbb{X})$.
- The key remains the maximization of the partial log-likelihood, independently of the chosen model for g .
- These extensions make it possible to explore richer forms of dependence and to improve predictive performance.

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